

Review of counting

1 (Distinguishability). How many ways are there to put

- (a) 4 distinguishable balls into 2 distinguishable boxes
- (b) 4 distinguishable balls into 2 indistinguishable boxes
- (c) 4 indistinguishable balls into 2 distinguishable boxes
- (d) 4 indistinguishable balls into 2 indistinguishable boxes

Solution. (a) We can view each ball has have to choose between either box, amounting to 2^4 total options.
(b) Divide (a) by the number of rearrangements of the boxes: $2^4/2 = 2^3$.
(c) Since we can't tell the balls apart, we only care about how many we put in each box. Picking how many to put in the first box determines how many go in the second (that is, the remainder of the balls must go in the second box). We can either put 0, 1, 2, 3 or 4 balls in the first box, meaning there's only 5 ways.
(d) As with (c) only the number in each box matters, but this time we have to be careful not to overcount. Our options are $\{4, 0\}$, $\{1, 3\}$, $\{2, 2\}$ so there are 3 ways.

□

2 (Bose-Einstein). Find the number of ways to put k indistinguishable balls into n distinguishable boxes

Solution. We employ the *stars and bars* technique. Consider $n + 1$ barriers instead of n boxes. The problem is then to construct a sequence of “stars” and “bars” (i.e., barriers) such that the sequence starts and ends with a bar, and contains any arrangement of stars and bars in between. Each such sequence corresponds to a placement of k indistinguishable balls into n boxes: reading from left to right, the number of stars between two bars is the number of balls in the corresponding box.

Each such sequence consists of $n + k + 1$ slots, the first and last must consist of bars. Therefore to construct a sequence we need to figure out where to place the stars among the $n + k - 1$ remaining slots. This amounts to

$$\binom{n + k - 1}{k}$$

many options, which is what we wanted to find.

□

Probability spaces

3. Can there be a uniform probability measure on the set of natural numbers?

Solution. No.

□

4. Some properties of the symmetric difference

- (a) Find an expression for $\mathbb{P}(A \Delta B) = \mathbb{P}(A) + \mathbb{P}(B) - 2\mathbb{P}(A \cap B)$.
- (b) Show that if $\mathbb{P}(A \Delta B) = 0$ then $\mathbb{P}(A) = \mathbb{P}(B)$.

Solution. The first part is evident by looking at the Venn diagram.

The second is a bit trickier. Notice that $\mathbb{P}(A \Delta B) = \mathbb{P}(A - B) + \mathbb{P}(B - A)$ which means that if $\mathbb{P}(A \Delta B) = 0$ then $\mathbb{P}(A - B) = \mathbb{P}(B - A) = 0$. Moreover, since $A = (A - B) \cup (A \cap B)$ and likewise $B = (B - A) \cup (A \cap B)$, then

$$\mathbb{P}(A) = \mathbb{P}(A - B) + \mathbb{P}(A \cap B) = \mathbb{P}(A \cap B) = \mathbb{P}(B - A) + \mathbb{P}(A \cap B) = \mathbb{P}(B)$$

as wanted. □